

Weekly Progress Report - Week 7

Improve performance of existing online trackers for small object detection [UAV Videos]

Group: The Convolutioners

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Outline of Performed Tasks:

- Compiled all experimental results across all pipeline configurations and components into a unified comparison
- Analysed detection, Re-ID, and tracking results to draw conclusions about system performance and failure modes
- Identified key trade-offs between the YOLO+Siamese and SAHI+Siamese pipeline variants
- Documented limitations of the current system and proposed directions for future work
- Prepared the final project report covering all phases of the project
- Prepared presentation slides summarising the methodology, results, and insights

Result Compilation and Analysis

This final week was dedicated to bringing together all results produced across the project into a coherent and complete analysis. Experimental outputs from all five phases, data preparation, YOLO detection, SAHI tiling, Siamese Re-ID optimisation, and custom BoT-SORT tracking, were consolidated and examined jointly to assess the overall effectiveness of the proposed system.

Presentation and Final Report Preparation

The final project report was prepared covering all five phases of the project in full detail, including the dataset preparation procedure, baseline evaluations, SAHI tiling integration, Siamese Re-ID design and optimisation, and custom BoT-SORT tracker development and benchmarking. All experimental tables, figures, and metric comparisons were compiled and cross-referenced within the report.

Presentation slides were also prepared to provide a concise and accessible summary of the project for the final demonstration. The presentation covers the project motivation and problem statement, an overview of the proposed pipeline, key results from the detection and tracking benchmarks, and the conclusions and future work identified through the analysis.